INCH-POUND

MIL-DTL-32375D (MR)
21 April 2025
SUPERSEDING
MIL-DTL-32375C (MR)
26 June 2023

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM ALLOY, 7085, 7056, 7075, 7181, and 7099, UNWELDABLE APPLIQUE

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers AA7085, AA7056, and AA7181 wrought aluminum alloy armor plate for non-fusion welded applications in nominal thicknesses from 0.500 to 3.000 inch, inclusive, AA7075 wrought aluminum alloy armor plate for non-fusion welded applications in nominal thicknesses from 1.000 to 1.500 inch, inclusive, and AA7099 wrought aluminum alloy armor plate for non-fusion welded applications in nominal thicknesses from 2.000 to 3.000 inch, inclusive (see 6.2). Additional applications for the AA7056 alloy with a T741 temper is also included in this specification for ballistic plate with an ordered thickness of 2.500 to 3.000 inches. The fusion weldability of these wrought aluminum alloy armors has not been tested or analyzed over this nominal thickness range. Therefore, these alloys should only be used as appliqué armor until a weldability study has been completed with acceptable results.

1.2 Classification. The wrought aluminum armor should be of the following class and tempers (types), as specified (see 6.2).

1.2.1 Class I. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7085 aluminum alloy. The applicable gauge range for class I is 0.500-3.000 inches.

1.2.1.1 Type A. Class I, type A has a temper designation (See 6.4) which is a high-strength temper designed for maximum resistance to armor piercing (AP) projectiles.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9535

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1.2.1.2 Type B. Class I, type B has a temper designation (See 6.4) which is designed for maximum resistance to blast.

1.2.2 Class II. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7056 aluminum alloy. The applicable gauge range for class II is 0.500 - 3.000 inches.

1.2.2.1 Type A. Class II, type A has a temper designation (See 6.4) which is a high-strength temper designed for maximum resistance to armor piercing (AP) projectiles.

1.2.2.1.1 Type A1. Class II, type A1 has a temper designation (See 6.4) for special applications in the ordered thickness of 2.500 - 3.000 inches.

1.2.2.2 Type B. Class II, type B has a temper designation (See 6.4) which is designed for maximum resistance to blast.

1.2.3 Class III. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7075 aluminum alloy. The applicable gauge range for class III is 1.000 - 1.500 inches.

1.2.4 Class IV. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7181 aluminum alloy. The applicable gauge range for class IV is 0.500 - 3.000 inches.

1.2.4.1 Type A. Class IV, type A has a temper designation (See 6.4) which is a high-strength temper designed for maximum resistance to armor piercing (AP) projectiles.

1.2.4.2 Type B. Class IV, type B has a temper designation (See 6.4) which is designed for maximum resistance to blast.

1.2.5 Class V. Wrought aluminum armor that conforms to the Aluminum Association designation for the 7099 aluminum alloy. The applicable gauge range for class V is 2.000 - 3.000 inches.

1.2.5.1 Type B. Class V, type B has a temper designation (See 6.4) which is designed for maximum resistance to blast.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32505	-	Armor Plate, Aluminum, Alloys 7017 And 7020, Weldable

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V50 Ballistic Test for Armor
MIL-STD-3057	-	Arc Welding of Armor Grade Aluminum

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE ALUMINUM ASSOCIATION, INC.

Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org).

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products

ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products Spectrometry (Performance Based Method)
ASTM E716	-	Standard Practices for Sampling Aluminum and Aluminum Alloys for Spectrochemical Analysis
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Atomic Emission Spectrometry
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission
ASTM G38	-	Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens

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ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products

(Copies of these documents are available from https://www.astm.org).

SAE INTERNATIONAL

AMS 2750	-	Pyrometry
AMS 2772	-	Heat Treatment of Aluminum Alloy Raw Materials

(Copies of these documents are available from https://www.sae.org).

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required, and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical compositions for classes I, II, III, IV, and V alloys defined by the Aluminum Association Teal Sheets, “International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys” are listed in table I and shall be within those limits. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

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TABLE I. Chemical composition, weight percent. 1/ 2/

ELEMENTS	SYMBOL	Class I 7085 ALLOY	Class II 7056 ALLOY	Class III 7075 ALLOY	Class IV 7181 ALLOY	Class V 7099 ALLOY
Silicon	Si	0.06	0.10	0.4	0.08	0.12
Iron	Fe	0.08	0.12	0.5	0.10	0.15
Copper	Cu	1.3 - 2.0	1.2 - 1.9	1.2 - 2.0	1.2 - 1.9	1.4 - 2.1
Manganese	Mn	0.04	0.20	0.3	0.15	0.04
Magnesium	Mg	1.2 - 1.8	1.5 - 2.3	2.1 - 2.9	1.7 - 2.2	1.6 - 2.3
Chromium	Cr	0.04	---	0.18 - 0.28	0.04	0.04
Zinc	Zn	7.0 - 8.0	8.5 - 9.7	5.1 - 6.1	6.7 - 7.9	7.4 - 8.4
Titanium	Ti	0.06	0.08	0.2	0.06	0.06
Zirconium	Zr	0.08 - 0.15	0.05 - 0.15	---	0.08 - 0.18	0.05 - 0.15
Other, max. Each	---	0.05	0.05	0.05	0.05	0.05
Other, max. Total 3/	---	0.15	0.15	0.15	0.15	0.15
Aluminum	Al	Remainder	Remainder	Remainder	Remainder	Remainder

1/ Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those "others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.3 Mechanical properties. Unless otherwise specified in the contract or purchase order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal transverse (LT) direction shall meet the minimum mechanical properties listed in table IIa, table IIb, table IIc, and/or table IId. If mechanical property requirements differ from those contained in table IIa, table IIb, table IIc, or table IId or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

TABLE IIa. Minimum mechanical properties for class I, AA7085. 1/, 2/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Type A	Type B	Type A	Type B	Type A	Type B
0.500 to 1.500, incl.	80	68	74	60	11	12
1.501 to 2.000, incl.	78	67	73	59	11	12
2.001 to 3.000, incl.	77	67	72	58	10	11

1/ The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.
2/ Values are taken in the Longitudinal Transverse Direction (LT).

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TABLE IIb. Minimum mechanical properties for class II, AA7056. [1/, 2/]

Thickness, inches	Tensile Strength, ksi			Yield Strength, 0.2% Offset, ksi			Elongation percent		
	Type A	Type A1	Type B	Type A	Type A1	Type B	Type A	Type A1	Type B
0.500 to 0.999, incl.	TBD [3/]	N/A [4/]	TBD [3/]	TBD [3/]	N/A [4/]	TBD [3/]	TBD [3/]	N/A [4/]	TBD [3/]
1.000 to 2.000, incl.	80	N/A [4/]	64	77	N/A [4/]	56	7	N/A [4/]	11
2.001 to 3.000, incl.	80	78 [5/]	64	77	74 [5/]	56	7	8 [5/]	11

[1/] The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.
[2/] Values are taken in the Longitudinal Transverse Direction (LT).
[3/] If the contract or purchase order requires plate in the ordered thickness of 0.500 to 1.000, incl. the minimum mechanical properties shall be listed in the contract or purchase order (see 6.2).
[4/] Not Applicable
[5/] Ordered Thickness is between 2.500 and 3.000 inches

TABLE IIc. Minimum mechanical properties for class III, AA 7075. [1/]

Thickness, inches	Tensile Strength, ksi	Yield Strength, 0.2% Offset, ksi	Elongation percent
1.000 to 1.500, incl.	70	60	9

[1/] Values are taken in the Longitudinal Transverse Direction (LT).

TABLE IId. Minimum mechanical properties for class IV, AA 7181. [1/, 2/]

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Type A	Type B	Type A	Type B	Type A	Type B
0.500 to 1.500, incl.	81	69	74	60	11	12
1.501 to 2.000, incl.	79	68	73	59	10	12
2.001 to 3.000, incl.	78	68	72	58	9	11

[1/] The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.
[2/] Values are taken in the Longitudinal Transverse Direction (LT).

TABLE IIe. Minimum mechanical properties for class V, AA7099. [1/]

Thickness, inches	Tensile Strength, ksi	Yield Strength, 0.2% Offset, ksi	Elongation percent
2.000 to 2.000, incl.	69	59	13
2.001 to 3.000, incl.	68	58	12

[1/] Values are taken in the Longitudinal Transverse Direction (LT).

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3.4 Weldability. The 7000 series aluminum alloys qualified in this specification are not fusion weldable and are solely for applique purposes (see 1.1 and 6.1). When a vehicle requires weldments, the fabricators (OEMs and kit manufacturers, etc.) should refer to qualified 7000 series alloys with demonstrated weldability in accordance with MIL-STD-3057, "Arc Welding of Armor Grade Aluminum" covered under MIL-DTL-32505.

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.6 Stress corrosion resistance. Stress Corrosion Cracking resistance is an optional requirement under this specification. When stress corrosion testing is requested in the contract or purchase order (see 6.2) and unless otherwise specified in the contract or purchase order (see 6.2), plate 0.750 inch and over in thickness shall be resistant to stress corrosion cracking, with a minimum of 5 of the 9 specimens showing no evidence of cracking at the end of 96 hours when tested as specified in 4.7.4.

3.7 Thermal processing. Heat treatment shall conform to the requirements of SAE AMS 2750 and SAE AMS 2772 and shall be such as to enable the material to meet the requirements of these specifications.

3.8 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or purchase order (see 6.2).

3.8.1 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2, except for thickness.

3.8.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in Table III.

TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		SPECIFIED WIDTH (Inches)								
		OVER	0.00	39.38	59.07	78.75	98.44	118.12	137.81	157.49
		THRU	39.37	59.06	78.74	98.43	118.11	137.80	157.48	177.17
OVER	THRU	TOLERANCES - INCHES (PLUS and MINUS)								
0.500	1.000		0.031	0.031	0.037	0.043	0.051	0.060	0.070	0.085
1.000	1.575		0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.575	2.362		0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.362	3.000		0.075	0.075	0.085	0.100	0.105	0.125	---	---

3.9 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the

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lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall contain no more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.9.1 Ballistic test plates. In addition to the markings in 3.8, each ballistic test plate shall be marked with the letters PRE for First Article test plates and ACC for Acceptance test plates. This marking shall be impression stamped on the edge in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see A.5.2).

3.10 Ballistic test plate information. For each lot of aluminum alloy armor, a properly completed Aluminum Armor Test Data Form (see figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.11 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth, and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations, and other defects which may affect its use. Minor scratches, gouges, dents and similar damage (see 3.11.2) are acceptable if the thickness is not reduced below the minimum material tolerance thickness.

3.11.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

3.11.2 Imperfections. The depth of surface scratches, nicks, pits, or gouges shall not reduce the plate thickness below the allowable minimum thickness. Isolated individual pits over 0.015 inches deep but not over 0.030 inches deep and not within 6 inches of each other and do not violate the minimum allowable thickness, are acceptable.

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY ARMOR									
SPECIFICATION: MIL-DTL-32375			**REVISION: D**			**AMENDMENT:**			
FIRING RECORD:		**DATE:**		**TEST CONTRACT NO:**					
	PLATE MANUFACTURER / PRODUCER			**PRIME CONTRACTOR**					
Name:									
Address:									
POC:									
Phone No:									
Email:									
TEST ITEM IDENTIFICATION:				Ordered Thickness:					
Lot No. _____ **Plate No.** _____			**Alloy:** _____ **Temper:** _____			**Class:** _____ **Type:** _____			
PURPOSE: ___ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing									
SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample _____)									
CHEMISTRY ANALYSIS:	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Zr**
V	**Li**	**Ag**	**Ni**	**Mn+Cr**	**Zr+Ti**	**Other - Each**			**Other - Total**	**Al**
										Rem.
MECHANICAL PROPERTIES:									
UTS (ksi):			0.2% YS (ksi):			Elongation (%):			
BALLISTIC TEST RESULTS: **SEPARATELY HEAT TREATED TEST PLATE : YES ___ NO ___**									
	Obl. (deg)	**Actual Thickness (in)**	**Required V_{50} (fps)**	**Actual V_{50} (fps)**	**Pass/ Fail**				
Test:									
Projectile:									
	NOTES:								
SPALL:	Spall Size:		Remarks:						
DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT & SUPPLIER OFFICIALS									
LOT: [____ met] [____ failed to meet] the ballistic requirements of specification MIL-DTL-46027									
Government Representative:		Date:	Supplier Representative:		Date:				

FIGURE 1. Aluminum Armor Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Production inspection (see 4.4).

4.2 Lot. A lot shall consist of all plate(s) of the same alloy and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate(s) in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit. All lots shall be fully traceable back to the ingot source by the supplier.

4.3 First article inspection. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use in production.

4.3.1 First article inspection. First article inspection except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examinations of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One (1) tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two (2) plates, 12 inches by 36 inches of each thickness to be supplied on the contract or purchase order, shall be submitted for ballistic testing in accordance with Appendix A. The orientation of these plates with respect to the rolling direction shall be at the option of the producer.

4.5.1.4 Stress corrosion tests. When required (see 6.2), nine (9) stress corrosion test specimens shall be removed from the same plate that has been selected for ballistic testing. The location and type of specimen shall be as specified in 4.7.4.

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4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one (1) sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one (1) sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.3 Ballistic testing. One (1) plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.3.

4.5.2.4 Stress corrosion tests. When required (see 6.2), nine test specimens shall be prepared from one plate randomly selected from each lot for stress corrosion testing. The plate shall meet the requirements when tested as specified in 4.7.4.

4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.8) and workmanship (see 3.11) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.8 in accordance with the sampling procedures approved by the procuring activity (see 6.2). At a minimum, one plate per lot shall be randomly selected for dimensional inspection.

4.7 Test specimens.

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4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E716, E1251 and E3061. In case of any dispute, analysis by method E3061 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal transverse (LT) direction. For material less than 0.500 inch in thickness, a standard rectangular tension test specimen shall be used. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Weldability. The alloys qualified in this specification are solely for applique purposes and are not subject to verification for fusion weldability (see 1.1, 3.4, and 6.1).

4.7.4 Ballistic testing. The ordered thickness specified in the contract or purchase order shall be used to determine the test projectile in accordance with table V. Ballistic testing shall be in accordance with Appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with table V and Appendix A to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL(P) for that plate.

TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees	CLASS I, II, & IV TYPE A	CLASS I, II, IV & V TYPE B	CLASS III
0.500 - 0.749	Cal. .30 AP M2	30	Table A-I	Table A-I	N/A
0.750 - 0.950	Cal. .50 FSP	0	Table A-II	Table A-II	N/A
0.951 - 1.500 class I&II 1.000 - 1.500 class III	20-mm FSP	0	Table A-III	Table A-III	Table A-III
0.750 - 1.500 class I&II 1.000 - 1.500 class III	Cal. .30 AP M2	0	Table A-IV	Table A-IV	Table A-VII
1.501 - 2.499	Cal. 50 AP M2	0	Table A-V	Table A-V	N/A
2.500 - 3.000	Cal. 50 AP M2	0	N/A	Table A-V	N/A
2.500 - 3.000	14.5-mm BS-41	0	Table A-VI	N/A	N/A

4.7.4.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

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4.7.4.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

4.7.5 Stress corrosion specimens. When stress corrosion testing is required (see 6.2), Short transverse stress corrosion test specimens shall be prepared in accordance with ASTM G38 and tested in accordance with ASTM G47. Nine specimens shall be tested per sample per lot. The specimens shall be stressed to 45 ksi. A maximum delay of 3 hours between stressing and initiation of stress corrosion test is permitted.

4.8 Rejection and retest. Unless otherwise specified in the contract or purchase order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, 4.8.2, and 4.8.3.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails, the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic retest. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.8.3 Stress corrosion retest. Immediately upon notification of the failure of the stress corrosion specimens, the manufacturer may at their expense submit thirteen additional specimens for retesting. A minimum of seven specimens out of the thirteen additional specimens shall show no cracks upon examination at the end of 96 hours when tested as specified in 4.7.4. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2, 4.7.3 and 4.7.4.

4.9 Reduced testing. At the discretion of the procuring activity (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing the manufacturer agrees to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained

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by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Armor classes specified herein as type A are intended for lightweight applications where resistance to ball and armor piercing types of ammunition and multiple hit capabilities are required. Armor classes specified herein as type B are intended for use on combat and tactical vehicles to protect the occupants from underbody threats (blast). Class II, type A1 is intended to be used on combat vehicles with a thickness of 2.500 to 3.000 inches. Class III armor is intended for use as an applique (B-Kit) for protection against Improvised Explosive Devices (IEDs) and armor piercing (AP) threats.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Specify classification of alloy, and ordered thickness, and type (see 1.1)
(c) When first article is required (see 3.1).
(d) Special mechanical properties and ballistic requirements, if required (see 3.3 and table 2b, footnote 3).
(e) When stress corrosion testing is required and if the test conditions are different (see 3.6, 4.5.1.4, 4.5.2.4, and 4.7.5).
(f) Dimension and tolerance requirements if other than in 3.8 and 3.8.1.
(g) If markings are different (see 3.8).
(i) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(j) The orientation of the ballistic plate is different (see 4.5.1.3 and 4.5.2.3).
(k) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(l) If approval was requested and received for a different ballistic testing facility (see 4.7.4.1).
(m) Rejection and retest requirement, if other than in 4.8.
(n) If reduced testing is allowed (see 4.9).
(o) Packaging requirements (see 5.1).

6.3 Density. The density of the specified alloys listed in this specification are as follows:

AA7085 = 0.103 lb/in^3 (2.85 g/cc)
AA7056 = 0.104 lb/in^3 (2.87 g/cc)
AA7075 = 0.101 lb/in^3 (2.81 g/cc)
AA7181 = 0.102 lb/in^3 (2.84 g/cc)
AA7099 = 0.102 lb/in^3 (2.84 g/cc)

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6.4 Tempers. The following tempers have shown to produce acceptable material when used to obtain the desired condition; maximum resistance to armor piercing (AP) projectiles, or maximum resistance to blast.

For class I, type A - temper of T711
For class I, type B - temper of T721

For class II, type A - temper of T761
For class II, type A1 - temper of T741 (for special applications)
For class II, type B - temper of T721

For class IV, type A - temper of T761
For class IV, type B - temper of T721

For class V, type B - temper of T731

6.5 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.6 Subject term (key word) listing.

Ballistic testing
Caliber .30 AP M2
Caliber .50 AP M2
Caliber .50 FSP
Military vehicles
20-mm FSP

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

BALLISTIC TESTING OF ARMOR PLATE, ALUMINUM ALLOY, 7085, 7056, 7075, and 7181, UNWELDABLE APPLIQUE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, un-weldable, 7085, 7056, 7075, 7181, and 7099 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification, and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/).

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any impact which is not a complete penetration may be considered a partial penetration.

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APPENDIX A

A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-VII.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the

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high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at his expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification including the plates utilized for acceptance retests are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in A.5.3.1.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits for class I, II, and IV - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type A	Type B		Type A	Type B
0.415	1424	1355	0.525	1699	1578
0.420	1437	1366	0.530	1711	1588
0.425	1451	1376	0.535	1722	1597
0.430	1464	1387	0.540	1733	1606
0.435	1478	1398	0.545	1745	1616
0.440	1491	1409	0.550	1756	1625
0.445	1504	1419	0.555	1767	1634
0.450	1517	1430	0.560	1778	1643
0.455	1530	1440	0.565	1789	1652
0.460	1542	1450	0.570	1800	1661
0.465	1555	1460	0.575	1811	1670
0.470	1567	1471	0.580	1822	1679
0.475	1580	1481	0.585	1832	1688
0.480	1592	1491	0.590	1843	1697
0.485	1604	1501	0.595	1854	1705
0.490	1617	1511	0.600	1864	1714
0.495	1629	1520	0.605	1875	1723
0.500 1/	1641	1530	0.610	1885	1731
0.505	1653	1540	0.615	1895	1740
0.510	1664	1550	0.620	1906	1748
0.515	1676	1559	0.625	1916	1757
0.520	1688	1569	0.630	1926	1765

1/ Specification type A and B requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits for class I, II, and IV - caliber .30 AP M2 projectiles at 30° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type A	Type B		Type A	Type B
0.635	1936	1774	0.740	2138	1942
0.640	1946	1782	0.745	2147	1950
0.645	1956	1791	0.749 2/	2154	1956
0.650	1966	1799	0.750	2156	1958
0.655	1976	1807	0.755	2165	1965
0.660	1986	1815	0.760	2174	1973
0.665	1996	1824	0.765	2183	1980
0.670	2006	1832	0.770	2192	1988
0.675	2015	1840	0.775	2201	1995
0.680	2025	1848	0.780	2210	2003
0.685	2035	1856	0.785	2219	2010
0.690	2044	1864	0.790	2228	2018
0.695	2054	1872	0.795	2236	2025
0.700	2063	1880	0.800	2245	2032
0.705	2073	1888	0.805	2254	2040
0.710	2082	1896	0.810	2262	2047
0.715	2092	1904	0.815	2271	2054
0.720	2101	1911	0.820	2280	2062
0.725	2110	1919	0.825	2288	2069
0.730	2120	1927	0.830	2297	2076
0.735	2129	1935	0.835	2305	2083

2/ Specification type A and B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits for class I, II, and IV - caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type A	Type B		Type A	Type B
0.665	1673	1638	0.760	2039	1969
0.670	1690	1654	0.765	2061	1988
0.675	1708	1670	0.770	2082	2008
0.680	1726	1686	0.775	2104	2027
0.685	1744	1703	0.780	2126	2047
0.690	1762	1719	0.785	2148	2067
0.695	1781	1736	0.790	2171	2087
0.700	1800	1753	0.795	2193	2107
0.705	1818	1770	0.800	2216	2128
0.710	1837	1787	0.805	2240	2149
0.715	1857	1805	0.810	2263	2169
0.720	1876	1822	0.815	2287	2191
0.725	1896	1840	0.820	2311	2212
0.730	1916	1858	0.825	2335	2234
0.735	1936	1876	0.830	2359	2255
0.740	1956	1894	0.835	2384	2277
0.745	1976	1913	0.840	2409	2299
0.750 1/	1997	1931	0.845	2434	2322
0.755	2018	1950	0.850	2460	2344

1/ Specification type A and B requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits for class I, II, and IV - caliber .50 fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Type A	Type B		Type A	Type B
0.855	2486	2367	0.950 2/	3030	2846
0.860	2512	2390	0.955	3061	2874
0.865	2538	2414	0.960	3093	2902
0.870	2565	2437	0.965	3126	2930
0.875	2591	2461	0.970	3159	2959
0.880	2619	2485	0.975	3192	2987
0.885	2646	2509	0.980	3225	3016
0.890	2674	2533	0.985	3259	3046
0.895	2702	2558	0.990	3293	3075
0.900	2730	2583	0.995	3328	3105
0.905	2759	2608	1.000	3362	3136
0.910	2787	2634	1.005	3398	3166
0.915	2817	2659	1.010	3433	3197
0.920	2846	2685	1.015	3469	3228
0.925	2876	2711	1.020	3505	3260
0.930	2906	2738	1.025	3542	3291
0.935	2936	2764	1.030	3579	3323
0.940	2967	2791	1.035	3617	3356
0.945	2998	2819	1.040	---	---

2/ Specification type A and B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits for class I, II, III, and IV - 20 mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A Class III	Class I, II, & IV Type B		Class I, II, & IV Type A Class III	Class I, II, & IV Type B
0.865	1090	1154	1.000 [2/]	1326	1373
0.870	1098	1162	1.005	1335	1382
0.875	1106	1169	1.010	1345	1391
0.880	1114	1177	1.015	1355	1400
0.885	1122	1185	1.020	1365	1409
0.890	1130	1192	1.025	1375	1418
0.895	1139	1200	1.030	1385	1427
0.900	1147	1208	1.035	1395	1436
0.905	1155	1215	1.040	1405	1445
0.910	1164	1223	1.045	1415	1455
0.915	1172	1231	1.050	1425	1464
0.920	1181	1239	1.055	1436	1473
0.925	1189	1247	1.060	1446	1483
0.930	1198	1255	1.065	1457	1492
0.935	1207	1263	1.070	1467	1502
0.940	1215	1271	1.075	1478	1512
0.945	1224	1279	1.080	1489	1521
0.950	1233	1288	1.085	1500	1531
0.951 [1/]	1235	1289	1.090	1511	1541
0.955	1242	1296	1.095	1522	1551
0.960	1251	1304	1.100	1533	1561
0.965	1260	1313	1.105	1544	1571
0.970	1269	1321	1.110	1555	1581
0.975	1279	1330	1.115	1566	1591
0.980	1288	1338	1.120	1578	1602
0.985	1297	1347	1.125	1589	1612
0.990	1307	1355	1.130	1601	1622
0.995	1316	1364	1.135	1613	1633

[1/] Specification class I, II, and IV; type A and B requirements begin for this ordered thickness.
[2/] Specification class III requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits for class I, II, III, and IV - 20 mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A Class III	Class I, II, & IV Type B		Class I, II, & IV Type A Class III	Class I, II, & IV Type B
1.140	1624	1643	1.295	2034	2005
1.145	1636	1654	1.300	2048	2018
1.150	1648	1664	1.305	2063	2031
1.155	1660	1675	1.310	2078	2044
1.160	1672	1686	1.315	2094	2057
1.165	1684	1697	1.320	2109	2070
1.170	1696	1708	1.325	2124	2084
1.175	1709	1719	1.330	2140	2097
1.180	1721	1730	1.335	2155	2111
1.185	1734	1741	1.340	2171	2124
1.190	1746	1752	1.345	2187	2138
1.195	1759	1763	1.350	2203	2152
1.200	1772	1775	1.355	2219	2166
1.205	1785	1786	1.360	2235	2179
1.210	1798	1798	1.365	2251	2194
1.215	1811	1809	1.370	2267	2208
1.220	1824	1821	1.375	2284	2222
1.225	1837	1833	1.380	2300	2236
1.230	1851	1844	1.385	2317	2251
1.235	1864	1856	1.390	2334	2265
1.240	1878	1868	1.395	2351	2280
1.245	1891	1880	1.400	2368	2294
1.250	1905	1892	1.405	2385	2309
1.255	1919	1905	1.410	2403	2324
1.260	1933	1917	1.415	2420	2339
1.265	1947	1929	1.420	2438	2354
1.270	1961	1942	1.425	2456	2369
1.275	1976	1954	1.430	2473	2384
1.280	1990	1967	1.435	2491	2400
1.285	2004	1979	1.440	2510	2415
1.290	2019	1992	1.445	2528	2431

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APPENDIX A

TABLE A-III. Minimum required ballistic limits for class I, II, III, and IV - 20 mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A Class III	Class I, II, & IV Type B		Class I, II, & IV Type A Class III	Class I, II, & IV Type B
1.450	2546	2446	1.530	2860	2711
1.455	2565	2462	1.535	2880	2728
1.460	2583	2478	1.540	2901	2746
1.465	2602	2494	1.545	2922	2764
1.470	2621	2510	1.550	2944	2782
1.475	2640	2526	1.555	2965	2799
1.480	2660	2542	1.560	2987	2817
1.485	2679	2559	1.565	3008	2836
1.490	2698	2575	1.570	3030	2854
1.495	2718	2592	1.575	3052	2872
1.500 [3/]	2738	2609	1.580	3075	2891
1.505	2758	2625	1.585	3097	2909
1.510	2778	2642	1.590	3120	2928
1.515	2798	2659	1.595	3142	2947
1.520	2818	2676	1.600	3165	2966
1.525	2839	2694	1.605	3188	2985

[3/] Specification class I, II, and IV; type A and B, and class III requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits for class I, II, and IV, types A and B, and class III - caliber .30 APM2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Type A	Type B	Class III		Type A	Type B	Class III
0.665	1800	1686	---	0.820	2057	1917	---
0.670	1809	1694	---	0.825	2064	1924	---
0.675	1818	1702	---	0.830	2072	1931	---
0.680	1826	1710	---	0.835	2080	1938	---
0.685	1835	1717	---	0.840	2087	1945	---
0.690	1844	1725	---	0.845	2095	1952	---
0.695	1852	1733	---	0.850	2103	1959	---
0.700	1861	1741	---	0.855	2110	1966	---
0.705	1870	1749	---	0.860	2118	1973	---
0.710	1878	1756	---	0.865	2125	1979	---
0.715	1887	1764	---	0.870	2133	1986	---
0.720	1895	1771	---	0.875	2140	1993	---
0.725	1903	1779	---	0.880	2148	2000	---
0.730	1912	1787	---	0.885	2155	2006	---
0.735	1920	1794	---	0.890	2162	2013	---
0.740	1928	1802	---	0.895	2170	2020	---
0.745	1937	1809	---	0.900	2177	2026	---
0.750 1/	1945	1816	---	0.905	2185	2033	---
0.755	1953	1824	---	0.910	2192	2040	---
0.760	1961	1831	---	0.915	2199	2046	2095
0.765	1969	1839	---	0.920	2206	2053	2102
0.770	1977	1846	---	0.925	2214	2059	2108
0.775	1985	1853	---	0.930	2221	2066	2114
0.780	1994	1860	---	0.935	2228	2072	2121
0.785	2002	1868	---	0.940	2235	2079	2127
0.790	2009	1875	---	0.945	2242	2085	2133
0.795	2017	1882	---	0.950	2249	2092	2139
0.800	2025	1889	---	0.955	2256	2098	2146
0.805	2033	1896	---	0.960	2263	2105	2152
0.810	2041	1903	---	0.965	2270	2111	2158
0.815	2049	1910	---	0.970	2278	2117	2165

1/ Specification type A and B requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits for class I, II, and IV - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Type A	Type B	Class III		Type A	Type B	Class III
0.975 2/	2284	2124	2171	1.130	2492	2312	2367
0.980	2291	2130	2177	1.135	2498	2318	2373
0.985	2298	2136	2184	1.140	2504	2323	2379
0.990	2305	2143	2190	1.145	2511	2329	2386
0.995	2312	2149	2196	1.150	2517	2335	2392
1.000	2319	2155	2202	1.155	2523	2341	2398
1.005	2326	2161	2209	1.160	2530	2346	2405
1.010	2333	2168	2215	1.165	2536	2352	2411
1.015	2340	2174	2221	1.170	2542	2358	2417
1.020	2347	2180	2228	1.175	2549	2364	2423
1.025	2353	2186	2234	1.180	2555	2369	2430
1.030	2360	2192	2240	1.185	2561	2375	2436
1.035	2367	2198	2247	1.190	2567	2381	2442
1.040	2374	2205	2253	1.195	2574	2386	2449
1.045	2380	2211	2259	1.200	2580	2392	2455
1.050	2387	2217	2266	1.205	2586	2397	2461
1.055	2394	2223	2272	1.210	2592	2403	2468
1.060	2400	2229	2278	1.215	2598	2409	2474
1.065	2407	2235	2284	1.220	2604	2414	2480
1.070	2414	2241	2291	1.225	2611	2420	2487
1.075	2420	2247	2297	1.230	2617	2425	2493
1.080	2427	2253	2303	1.235	2623	2431	2499
1.085	2433	2259	2310	1.240	2629	2436	2506
1.090	2440	2265	2316	1.245	2635	2442	2512
1.095	2446	2271	2322	1.250	2641	2447	2518
1.100	2453	2277	2329	1.255	2647	2453	2525
1.105	2459	2282	2335	1.260	2653	2458	2531
1.110	2466	2288	2341	1.265	2659	2464	2537
1.115	2472	2294	2348	1.270	2665	2469	2544
1.120	2479	2300	2354	1.275	2671	2475	2550
1.125	2485	2306	2360	1.280	2677	2480	2556

2/ Specification class III requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits for class I, II, and IV - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Type A	Type B	Class III		Type A	Type B	Class III
1.285	2683	2486	2563	1.450	2872	2658	2772
1.290	2689	2491	2569	1.455	2878	2663	2778
1.295	2695	2496	2575	1.460	2884	2668	2785
1.300	2701	2502	2582	1.465	2889	2673	2791
1.305	2707	2507	2588	1.470	2895	2678	2798
1.310	2712	2513	2594	1.475	2900	2683	2804
1.315	2718	2518	2601	1.480	2906	2688	2810
1.320	2724	2523	2607	1.485	2911	2693	2817
1.325	2730	2529	2613	1.490	2917	2698	2823
1.330	2736	2534	2620	1.495	2922	2703	2829
1.335	2742	2539	2626	1.500 [2/]	2928	2708	2836
1.340	2748	2544	2632	1.505	2933	2713	2842
1.345	2753	2550	2639	1.510	2938	2718	2848
1.350	2759	2555	2645	1.515	2944	2723	2855
1.355	2765	2560	2652	1.520	2949	2728	2861
1.360	2771	2566	2658	1.525	2955	2733	2868
1.365	2776	2571	2664	1.530	2860	2711	2874
1.370	2782	2576	2671	1.535	2880	2728	2880
1.375	2788	2581	2677	1.540	2901	2746	2887
1.380	2794	2586	2683	1.545	2922	2764	2893
1.385	2799	2592	2690	1.550	2944	2782	2899
1.390	2805	2597	2696	1.555	2965	2799	2906
1.395	2811	2602	2702	1.560	2987	2817	2912
1.400	2816	2607	2709	1.565	3008	2836	2918
1.405	2822	2612	2715	1.570	3030	2854	2925
1.410	2828	2617	2721	1.575	3052	2872	2931
1.415	2833	2623	2728	1.580	3075	2891	2938
1.420	2839	2628	2734	1.585	3097	2909	2944
1.425	2845	2633	2740	1.590	3120	2928	2950
1.430	2850	2638	2747	1.595	3142	2947	2957
1.435	2856	2643	2753	1.600	3165	2966	2963
1.440	2861	2648	2759	1.605	3188	2985	2969
1.445	2867	2653	2766	1.610	---	---	---

[2/] Specification class I, II, and IV; type A and B, and class III requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits for class I, II, IV, and V - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A	Class I, II, & IV Type B		Class I, II, & IV Type A	Class I, II, & IV Type B
1.395	1995	1824	1.545	2143	1968
1.400	2000	1829	1.550	2147	1973
1.405	2006	1834	1.555	2152	1977
1.410	2011	1839	1.560	2157	1982
1.415	2016	1844	1.565	2161	1987
1.420	2021	1849	1.570	2166	1991
1.425	2026	1854	1.575	2171	1996
1.430	2031	1859	1.580	2175	2000
1.435	2036	1864	1.585	2180	2005
1.440	2041	1869	1.590	2185	2009
1.445	2046	1873	1.595	2189	2014
1.450	2051	1878	1.600	2194	2018
1.455	2056	1883	1.605	2199	2023
1.460	2060	1888	1.610	2203	2027
1.465	2065	1893	1.615	2208	2032
1.470	2070	1898	1.620	2213	2036
1.475	2075	1902	1.625	2217	2041
1.480	2080	1907	1.630	2222	2045
1.485	2085	1912	1.635	2226	2050
1.490	2090	1917	1.640	2231	2054
1.495	2095	1921	1.645	2235	2059
1.500	2100	1926	1.650	2240	2063
1.501 1/	2100	1927	1.655	2244	2067
1.505	2104	1931	1.660	2249	2072
1.510	2109	1936	1.665	2253	2076
1.515	2114	1940	1.670	2258	2081
1.520	2119	1945	1.675	2262	2085
1.525	2124	1950	1.680	2267	2089
1.530	2128	1954	1.685	2271	2094
1.535	2133	1959	1.690	2276	2098
1.540	2138	1964	1.695	2280	2102

1/ Specification class I, II, and IV; type A and B requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits for class I, II, IV, and V - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A	Class I, II, & IV Type B		Class I, II, & IV Type A	Class I, II, IV, & V Type B
1.700	2285	2107	1.855	2419	2237
1.705	2289	2111	1.860	2423	2241
1.710	2294	2115	1.865	2427	2245
1.715	2298	2120	1.870	2431	2249
1.720	2302	2124	1.875	2435	2253
1.725	2307	2128	1.880	2439	2257
1.730	2311	2132	1.885	2444	2261
1.735	2316	2137	1.890	2448	2265
1.740	2320	2141	1.895	2452	2269
1.745	2324	2145	1.900	2456	2273
1.750	2329	2149	1.905	2460	2277
1.755	2333	2154	1.910	2464	2281
1.760	2337	2158	1.915	2468	2285
1.765	2342	2162	1.920	2472	2289
1.770	2346	2166	1.925	2477	2293
1.775	2350	2170	1.930	2481	2297
1.780	2355	2175	1.935	2485	2301
1.785	2359	2179	1.940	2489	2305
1.790	2363	2183	1.945	2493	2309
1.795	2368	2187	1.950	2497	2313
1.800	2372	2191	1.955	2501	2317
1.805	2376	2195	1.960	2505	2320
1.810	2380	2200	1.965	2509	2324
1.815	2385	2204	1.970	2513	2328
1.820	2389	2208	1.975	2517	2332
1.825	2393	2212	1.980	2521	2336
1.830	2397	2216	1.985	2525	2340
1.835	2402	2220	1.990	2529	2344
1.840	2406	2224	1.995	2533	2348
1.845	2410	2228	2.000 [2/]	2537	2352
1.850	2414	2232	2.005	2541	2355

[2/] Specification class V, type B requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits for class I, II, IV, and V - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A	Class I, II, IV, & V Type B		Class I, II, & IV Type A	Class I, II, IV, & V Type B
2.010	2545	2359	2.170	2670	2480
2.015	2549	2363	2.175	2674	2483
2.020	2553	2367	2.180	2677	2487
2.025	2557	2371	2.185	2681	2491
2.030	2561	2375	2.190	2685	2494
2.035	2565	2379	2.195	2689	2498
2.040	2569	2382	2.200	2693	2502
2.045	2573	2386	2.205	2696	2505
2.050	2577	2390	2.210	2700	2509
2.055	2581	2394	2.215	2704	2512
2.060	2585	2398	2.220	2708	2516
2.065	2589	2401	2.225	2711	2520
2.070	2593	2405	2.230	2715	2523
2.075	2597	2409	2.235	2719	2527
2.080	2601	2413	2.240	2723	2531
2.085	2604	2416	2.245	2726	2534
2.090	2608	2420	2.250	2730	2538
2.095	2612	2424	2.255	2734	2541
2.100	2616	2428	2.260	2737	2545
2.105	2620	2432	2.265	2741	2548
2.110	2624	2435	2.270	2745	2552
2.115	2628	2439	2.275	2749	2556
2.120	2632	2443	2.280	2752	2559
2.125	2635	2446	2.285	2756	2563
2.130	2639	2450	2.290	2760	2566
2.135	2643	2454	2.295	2763	2570
2.140	2647	2458	2.300	2767	2573
2.145	2651	2461	2.305	2771	2577
2.150	2655	2465	2.310	2774	2580
2.155	2658	2469	2.315	2778	2584
2.160	2662	2472	2.320	2782	2587
2.165	2666	2476	2.325	2785	2591

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APPENDIX A

TABLE A-V. Minimum required ballistic limits for class I, II, IV, and V - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A	Class I, II, IV, & V Type B		Class I, II, & IV Type A	Class I, II, IV, & V Type B
2.330	2789	2594	2.485	2899	2701
2.335	2793	2598	2.490	2903	2704
2.340	2796	2601	2.495	2906	2708
2.345	2800	2605	2.499 3/	2909	2710
2.350	2803	2608	2.500	2910	2710
2.355	2807	2612	2.505	2913	2714
2.360	2811	2615	2.510	2917	2718
2.365	2814	2619	2.515	2920	2721
2.370	2818	2622	2.520	2924	2724
2.375	2821	2626	2.525	2927	2728
2.380	2825	2629	2.530	2931	2731
2.385	2829	2633	2.535	2934	2734
2.390	2832	2636	2.540	2938	2738
2.395	2836	2640	2.545	2941	2741
2.400	2839	2643	2.550	2945	2744
2.405	2843	2647	2.555	2948	2748
2.410	2847	2650	2.560	2952	2751
2.415	2850	2653	2.565	2955	2754
2.420	2854	2657	2.570	2958	2758
2.425	2857	2660	2.575	2962	2761
2.430	2861	2664	2.580	2965	2764
2.435	2864	2667	2.585	2969	2768
2.440	2868	2671	2.590	2972	2771
2.445	2871	2674	2.595	2976	2774
2.450	2875	2677	2.600	2979	2777
2.455	2878	2681	2.605	2982	2781
2.460	2882	2684	2.610	2986	2784
2.465	2885	2687	2.615	2989	2787
2.470	2889	2691	2.620	2993	2790
2.475	2892	2694	2.625	2996	2794
2.480	2896	2698	2.630	---	2797

3/ Specification class I, II and IV, type A requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits for class I, II, IV, and V - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A	Class I, II, IV, & V Type B		Class I, II, & IV Type A	Class I, II, IV, & V Type B
2.635	---	2800	2.790	---	2899
2.640	---	2803	2.795	---	2902
2.645	---	2807	2.800	---	2906
2.650	---	2810	2.805	---	2909
2.655	---	2813	2.810	---	2912
2.660	---	2816	2.815	---	2915
2.665	---	2820	2.820	---	2918
2.670	---	2823	2.825	---	2921
2.675	---	2826	2.830	---	2924
2.680	---	2829	2.835	---	2927
2.685	---	2833	2.840	---	2930
2.690	---	2836	2.845	---	2934
2.695	---	2839	2.850	---	2937
2.700	---	2842	2.855	---	2940
2.705	---	2845	2.860	---	2943
2.710	---	2849	2.865	---	2946
2.715	---	2852	2.870	---	2949
2.720	---	2855	2.875	---	2952
2.725	---	2858	2.880	---	2955
2.730	---	2861	2.885	---	2958
2.735	---	2865	2.890	---	2961
2.740	---	2868	2.895	---	2964
2.745	---	2871	2.900	---	2967
2.750	---	2874	2.905	---	2971
2.755	---	2877	2.910	---	2974
2.760	---	2880	2.915	---	2977
2.765	---	2884	2.920	---	2980
2.770	---	2887	2.925	---	2983
2.775	---	2890	2.930	---	2986
2.780	---	2893	2.935	---	2989
2.785	---	2896	2.940	---	2992

4/ Specification class I, II, IV, and class V, type B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits for class I, II, IV, and V - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I, II, & IV Type A	Class I, II, IV, & V Type B		Class I, II, & IV Type A	Class I, II, IV, & V Type B
2.945	---	2995	3.035	---	3049
2.950	---	2998	3.040	---	3052
2.955	---	3001	3.045	---	3055
2.960	---	3004	3.050	---	3058
2.965	---	3007	3.055	---	3061
2.970	---	3010	3.060	---	3064
2.975	---	3013	3.065	---	3067
2.980	---	3016	3.070	---	3070
2.985	---	3019	3.075	---	3073
2.990	---	3022	3.080	---	3076
2.995	---	3025	3.085	---	3079
3.000 4/	---	3028	3.090	---	3082
3.005	---	3031	3.095	---	3085
3.010	---	3034	3.100	---	3088
3.015	---	3037	3.105	---	3091
3.020	---	3040	3.110	---	3094
3.025	---	3043	3.115	---	3097
3.030	---	3046	3.120	---	3099
3.035	---	3049	3.125	---	3102

4/ Specification class I, II, IV, and class V, type B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits for class I, II, and IV; types A and A1 14.5-mm BS-41 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I, II, & IV Type A & Type A1		Class I, II, & IV Type A & Type A1
2.375	2579	2.520	2676
2.380	2582	2.525	2680
2.385	2586	2.530	2683
2.390	2589	2.535	2686
2.395	2592	2.540	2689
2.400	2596	2.545	2693
2.405	2599	2.550	2696
2.410	2603	2.555	2699
2.415	2606	2.560	2703
2.420	2609	2.565	2706
2.425	2613	2.570	2709
2.430	2616	2.575	2712
2.435	2620	2.580	2716
2.440	2623	2.585	2719
2.445	2626	2.590	2722
2.450	2630	2.595	2725
2.455	2633	2.600	2729
2.460	2636	2.605	2732
2.465	2640	2.610	2735
2.470	2643	2.615	2738
2.475	2646	2.620	2742
2.480	2650	2.625	2745
2.485	2653	2.630	2748
2.490	2656	2.635	2751
2.495	2660	2.640	2754
2.500	2663	2.645	2758
2.501 1/	2664	2.650	2761
2.505	2666	2.655	2764
2.510	2670	2.660	2767
2.515	2673	2.665	2770

1/ Specification class I, II, and IV, type A, and class II type A1 requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits for class I, II, and IV; types A and A1 14.5-mm BS-41 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I, II, & IV Type A & Type A1		Class I, II, & IV Type A & Type A1
2.670	2774	2.820	2868
2.675	2777	2.825	2871
2.680	2780	2.830	2874
2.685	2783	2.835	2877
2.690	2786	2.840	2880
2.695	2789	2.845	2883
2.700	2793	2.850	2886
2.705	2796	2.855	2889
2.710	2799	2.860	2892
2.715	2802	2.865	2895
2.720	2805	2.870	2898
2.725	2808	2.875	2901
2.730	2812	2.880	2904
2.735	2815	2.885	2907
2.740	2818	2.890	2910
2.745	2821	2.895	2914
2.750	2824	2.900	2917
2.755	2827	2.905	2920
2.760	2830	2.910	2923
2.765	2834	2.915	2926
2.770	2837	2.920	2929
2.775	2840	2.925	2932
2.780	2843	2.930	2935
2.785	2846	2.935	2938
2.790	2849	2.940	2941
2.795	2852	2.945	2944
2.800	2855	2.950	2947
2.805	2858	2.955	2950
2.810	2861	2.960	2953
2.815	2865	2.965	2956

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits for class I, II, and IV; types A and A1 14.5-mm BS-41 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I, II, & IV Type A & Type A1		Class I, II, & IV Type A & Type A1
2.970	2959	3.050	3006
2.975	2962	3.055	3009
2.980	2965	3.060	3012
2.985	2968	3.065	3015
2.990	2971	3.070	3018
2.995	2974	3.075	3021
3.000 [2]	2977	3.080	3024
3.005	2980	3.085	3027
3.010	2982	3.090	3030
3.015	2985	3.095	3032
3.020	2988	3.100	3035
3.025	2991	3.105	3038
3.025	2991	3.110	3041
3.030	2994	3.115	3044
3.035	2997	3.120	3047
3.040	3000	3.125	3050
3.045	3003	3.125	3050

[2/] Specification class I, II, and IV; type A, and class II, type A1 requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY - MR

(Project 9535-2025-005)

Review activities:
Army - AR, AT, AV, TE

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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